Status	Finished
Started	Saturday, 17 October 2026, 9:46 AM
Completed	Saturday, 17 October 2026, 9:52 AM
Duration	6 mins 28 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list is unsorted
c. When the list is sorted ✓
d. When the list contains duplicate elements

Question **2**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. At the beginning of the list
c. In the entire list
d. In the left sublist ✓

Question **3**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Complexity

b. Selection

c. Bubble ✓

d. Insertion

Question **4**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist

b. The list is sorted

c. The search ends successfully ✓

d. The search continues in the right sublist

Question **5**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Linear

b. Binary ✓

c. Both (1) & (3)

d. Hash

Question **6**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Finding
b. Mining
c. Discovering
d. Searching ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99
b. 90 and 99 ✓
c. 90 and 94
d. 89 and 94

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It does not work with negative numbers
c. It can only be applied to large lists
d. It requires the list to be sorted ✓

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(log n) ✓
- b. O(n log n)
- c. O(1)
- d. O👎

Question **10**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. 1
- b. log n
- c. n ✓
- d. n/2

Question **11**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is the last element in the array
- b. When the item is somewhere in the middle of the array ✓
- c. When the item is not the array at all
- d. Item is the last element in the array or item is not there at all

Question **12**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the list is sorted
- b. When the search algorithm finishes
- c. When the element is found in the list ✓
- d. When the list contains duplicate elements

Question **13**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. n-1
c. log n
d. n ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and sorted
b. When the list is small and unsorted ✓
c. When the list is large and unsorted
d. When the list is large and sorted

Question **15**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist
b. At the beginning of the list
c. In the middle of the list
d. In the right sublist ✓